

4 to 20 mA Digital Transmitter Board

Alphasense Type A and B Toxic Gas Sensors



Alphasense 4-20mA digital transmitters offer the following features:

- Factory calibration, custom-set for immediate use
- Digital zero, sensitivity and temperature compensation
- Digital and 4-20mA output
- Biased or unbiased operation
- Calibration and range change

Alphasense 4-20mA digital transmitter boards provide a cost-effective way for Original Equipment Manufacturers to include 4-20mA gas sensor transmitters for fixed installation systems. The range of sensors are shown in Table 1. Transmitters are supplied with pre-calibrated sensors.

The 4-20mA output signal performance is as shown on individual sensor datasheets. Calibration and digital interface communication use the 2-wire power supply (HART-type communications).

Optional fitting kit and connector/leads are available on request. Please see Figure 2.

Table 1. Transmitter Board and Sensors

Gas	Sensor Type A	Max concentration (ppm)	Sensor Type B	Max concentration (ppm)
Carbon Monoxide	CO-AF	5,000	CO-BF	5,000
	CO-AE	10,000	CO-B1	5,000
	CO-AX	2,000	CO-BX	2,000
Hydrogen Sulfide	H2S-A1	100	H2S-B1	200
	H2S-AH	50	H2S-BH	50
	H2S-AE	2,000	H2S-BE	2,000
Sulfur Dioxide	SO2-AF	50	SO2-BF	100
	SO2-AE	2,000		
Nitrogen Dioxide	NO2-A1	20	NO2-B1	20
	NO2-AE	200		
Chlorine	CL2-A1	20	CL2-B1	20
Nitric Oxide*	NO-A1	250	NO-B1	250
	NO-AE	5,000		
Phosphine	PH3-A1	10	PH3-B1	10
			PH3-BE	2,000
Ethylene Oxide*	ETO-A1	100	ETO-B1	100
Hydrogen Cyanide	HCN-A1	100	HCN-B1	100
Hydrogen Chloride	HCL-A1	20	HCL-B1	20
Ammonia			NH3-B1	100

* Biased sensors require at least 12 hours to stabilise after first powered on. Specify required full scale gas concentration.

Do not exceed maximum gas concentration shown in table.

INPUT

Type:	3-electrode gas sensor: type A or type B
Maximum range:	-220uA to +320 uA
Minimum range:	± 3 uA
Sample rate:	500ms per sample
Thermal drift:	0.05 uA/°C
Power-off state:	Shorting FET connects working to reference electrode
Resolution:	11 nA
Bias:	Selectable: Off, +200 and +300mV (±10 mV tolerance)
Temperature compensation:	On-board temperature sensor. Range -30°C to 60°C ± 0.5°C. Software correction of Zero and Span.
Connection:	2-pin MOLEX plug (ref 22-27-2021)

OUTPUT

Type:	4 mA to 20 mA 2-wire loop powered
Output current range:	3.8 mA to 21.5 mA
Supply voltage:	10 to 30 VDC
Supply sensitivity:	< 0.03% 10 to 30 VDC
Loop ripple effect:	± 2 uA measure @ 1 volt RMS 50Hz supply ripple
Thermal drift:	± 0.2 uA /°C
Accuracy:	± 5 uA
Maximum loop resistance:	700 Ω @ 24VDC
Resolution:	0.75 uA
Sensor over-range:	> 21.5 mA
Protection:	Reverse connection and over-voltage protection
Connection:	2-Pin MOLEX plug (Ref. 22-27-2021)

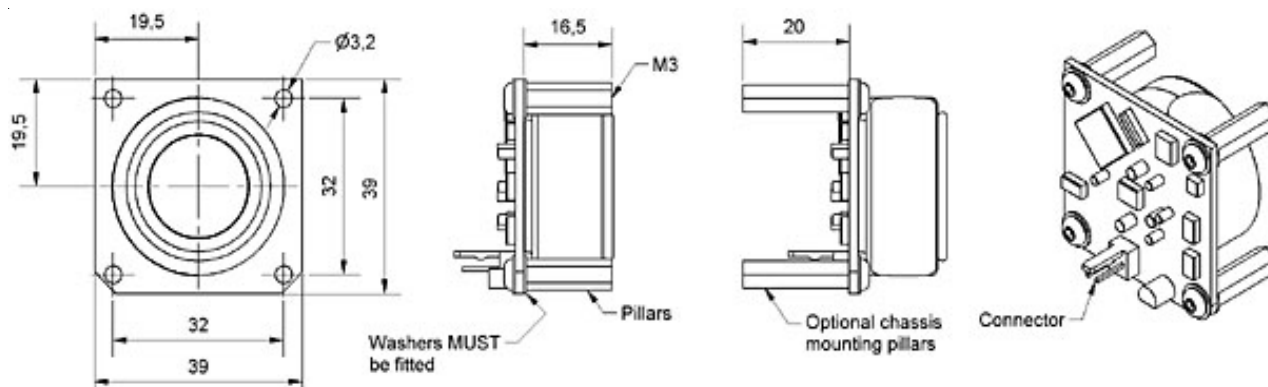
ENVIRONMENT

Ambient temperature:	-30°C to 60°C
Ambient storage:	-40°C to 70°C
Ambient humidity:	0% to 95% continuous (non-condensing)
Coating:	Conformal spray coated

CE APPROVAL

BS EN 61326 (Industrial)

Figure 2 Transmitter Board Dimension Details



Optional fitting kit # 000-0420-KIT

4 x pillars	16.0 mm length, M3 tapped
8 x washers	M3 flat washers
4 x screws	M3 x 8 button head screws
1 x transmitter lead	Molex socket housing with 150 mm leads

At the end of the product's life, do not dispose of any electronic sensor, component or instrument in the domestic waste, but contact the instrument manufacturer, Alphasense or its distributor for disposal instructions. NOTE: as applications of use are beyond our control, the information provided is given without legal responsibility. Customers should test under their own conditions, to ensure that the sensors and transmitters are suitable for their own requirements.

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